Collections of Data in C#

A collection of data means storing multiple values together so that they can be accessed and processed easily. In C#, arrays are one of the simplest ways to store a collection of values of the same data type.

1. Introduction to Arrays

An array is a collection of elements of the same data type stored under one variable name.

Syntax

```
data_type[] array_name = new data_type[size];
```

Example

```
using System;

class Program
{
   static void Main()
   {
      int[] marks = { 70, 80, 65, 90, 75 };


      Console.WriteLine("First Mark: " + marks[0]);
      Console.WriteLine("Third Mark: " + marks[2]);
   }
}
```

Using a loop

```
int[] numbers = { 10, 20, 30, 40, 50 };
for (int i = 0; i < numbers.Length; i++)
{
   Console.WriteLine(numbers[i]);
}
```

2. Multidimensional Arrays

A multidimensional array stores data in more than one dimension. The most common example is a 2D array, which represents rows and columns like a matrix.

Syntax

```
data_type[,] array_name = new data_type[rows, columns];
```

Example

```
using System;

class Program
{
   static void Main()
   {
      int[,] matrix =
      {
         { 10, 20, 30 },
         { 40, 50, 60 }
      };

      for (int i = 0; i < 2; i++)
      {
         for (int j = 0; j < 3; j++)
         {
            Console.Write(matrix[i, j] + " ");
         }

         Console.WriteLine();
      }
   }
}
```

3. Jagged Arrays

A jagged array is an array of arrays. Each row can have a different number of elements.

Example

```
using System;
```

```
class Program
{
   static void Main()
   {
      int[][] numbers = new int[3][];

      numbers[0] = new int[] { 10, 20 };
      numbers[1] = new int[] { 30, 40, 50 };
      numbers[2] = new int[] { 60, 70, 80, 90 };

      for (int i = 0; i < numbers.Length; i++)
      {
         for (int j = 0; j < numbers[i].Length; j++)
         {
            Console.Write(numbers[i][j] + " ");
         }

         Console.WriteLine();
      }
   }
}
```

4. Array Class

Length Returns total number of elements

Rank Returns number of dimensions

GetLength() Returns length of a specified dimension

Array.Sort() Sorts the elements

Array.Reverse() Reverses the elements

Array.IndexOf() Finds the index of an element

Array.Clear() Clears elements

Example

```
using System;

class Program
{
   static void Main()
   {
      int[] numbers = { 10, 20, 30, 40, 50 };

      Console.WriteLine("Length: " + numbers.Length);
      Console.WriteLine("Rank: " + numbers.Rank);
   }
}
```

5. Sorting an Array in C#

```
using System;
class Program
{
   static void Main()
   {
      int[] numbers = { 50, 20, 40, 10, 30 };

      Array.Sort(numbers);

      Console.WriteLine("Sorted Array:");

      foreach (int n in numbers)
      {
         Console.Write(n + " ");
      }
   }
```

```
}

using System;
class Program
{
   static void Main()
   {
      int[] numbers = { 50, 20, 40, 10, 30 };

      Array.Sort(numbers);
      Array.Reverse(numbers);

      Console.WriteLine("Descending Order:");

      foreach (int n in numbers)
      {
         Console.Write(n + " ");
      }
   }
}
```

//////////////////////////////////////

1. Introduction to String

A string can contain letters, numbers, spaces, and special characters.

Example

```
using System;

class Program
{
   static void Main()
   {
```

```
        string name = "Rahul";
        string city = "Mumbai";

        Console.WriteLine(name);
        Console.WriteLine(city);
    }
}
```

Accessing Characters

```
string name = "Hello";
Console.WriteLine(name[0]);
Console.WriteLine(name[1]);
```

2. String Formatting

String formatting is used to insert values into a string using placeholders such as {0}, {1}, etc.

Example

```
using System;
class Program
{
    static void Main()
    {
        string name = "Amit";
        int age = 22;

        string result = String.Format("My name is {0} and my age is {1}.", name, age);

        Console.WriteLine(result);
    }
}
```

5. StringBuilder

StringBuilder is used when we need to modify a string many times.

It belongs to the System.Text namespace.

Example

```
using System;
using System.Text;
class Program
{
   static void Main()
   {
      StringBuilder sb = new StringBuilder();


      sb.Append("Hello ");
      sb.Append("World");


      Console.WriteLine(sb);
   }
}
```

6. String Class

The String class belongs to the System namespace. It provides many methods for manipulating and working with strings.

Common String Methods

Method/Property Description

Length Returns number of characters

ToUpper() Converts to uppercase

ToLower() Converts to lowercase

Contains() Checks whether text exists

Substring() Extracts part of a string

Replace() Replaces text

Trim() Removes leading/trailing spaces

IndexOf() Finds the position of text

Example

```
using System;


class Program
{
   static void Main()
   {
      string text = "Hello C#";


      Console.WriteLine(text.Length);
      Console.WriteLine(text.ToUpper());
      Console.WriteLine(text.ToLower());
      Console.WriteLine(text.Contains("C#"));
      Console.WriteLine(text.Replace("C#", "Programming"));
   }
}
```